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Achieving Effective Water Control in Low-Viscosity Oil Reservoirs: A Case Study with RCP AICDs in Offshore China

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Abstract

Maintaining efficient oil production from horizontal wells with varying reservoir properties is a significant challenge. In such scenarios, water breakthroughs can occur prematurely, hindering hydrocarbon recovery. Traditional completion techniques often lack the ability to effectively address this heterogeneity and achieve optimal water control.

Autonomous Inflow Control Devices (AICDs) offer an innovative solution for mitigating water influx in horizontal wells. This downhole device automatically regulates fluid flow based on predefined parameters, enabling proactive water control and improved sweep efficiency. This paper presents a successful field case study utilizing RCP AICDs to control water production in a low-viscosity oil reservoir (1.47 cP) offshore China.

The paper explores not only the optimal design workflow and value justification for AICDs but also the challenges that led operators to embrace this technology. This paper also emphasizes that to achieve the best results from the use of such technologies, it is essential not only to ensure optimal design but also to operate the wells under conditions that allow the AICD to perform at its highest efficiency. Notably, when AICDs were operated under optimum conditions in this study, the water cut exhibited a continuous declining trend since then, indicating sustained and greater water control effectiveness.

This case study demonstrates the effectiveness of AICD technology in controlling water production as water cut reduction was observed at both the mid and later production stages in edge and bottom water reservoirs with low-viscosity crude oil. The well with AICD completion has managed, so far, to deliver 310% increase in cumulative oil production from the well compared to basecase/analogous wells while also extending the well life twice as much than expected. The well's actual cumulative oil production significantly surpassed pre-drilling predictions, reaching over 63,500 m³ and continuing.

The implemented RCP AICDs played a crucial role in enhancing water control, improved reservoir management and oil production throughout the well's lifecycle. By effectively controlling water influx in a reservoir with exceptionally low oil viscosity, this pushes the boundaries of AICD applicability. This study offers valuable insights into how these devices can improve field development strategies by mitigating water

production and reducing operating costs for lifting, processing and disposal of water plus greenhouse gas emission and energy consumption associated with the operations.

The Field Description

There is one oil layer (L layer) vertically developed in the LFX oilfield, which is an anticline structure oil reservoir with a unified oil-water interface of -2479.2m (TVDSS). It is a typical block shaped edge water sandstone oil reservoir. The average porosity of the core analysis is 20.3%, and the average permeability of the core is 1323mD, belonging to a medium porosity, medium to high permeability reservoir. The total number of oil wells in the oilfield is 13, with 11 open wells and daily oil production of 510m³/d. The overall water cut is 97.2%, and the recovery rate is 67.2%. The reserves of the main area of the oilfield (16.31 million cubic meters) have been fully utilized; In 2015, the reserves of the Northwest District were declared to control 157,700 cubic meters of original oil in place. As per this, well LH1, which was in a temporary abandoned state, was used to sidetrack a horizontal well for improving the well network in the northwest area of the oilfield and increasing the degree of reserve utilization and delaying production decline.

LH1 is a horizontal well located in the northwest area of the oilfield with reservoir formation of L. The depth of the oil reservoir is 2444.2 m to 2478.9 m; the height of the oil-bearing interval is 34.7 m with the effective thickness of the oil reservoir is 0.3 m to 22.1 m. The L layer is stable on the plane and is a set of marine shore sandbars sediments, mainly composed of fine to coarse grained feldspar quartz sandstone and feldspar sandstone, with a small amount of lithic feldspar sandstone. The reservoir lithology is single, mainly composed of feldspar quartz sandstone, and the storage space type is porous. The heterogeneity of the L layer is vertically strong, with a set of mudstone interlayers developed in the lower part of the reservoir and two sets of thin mudstone interlayers locally developed in the upper pure sandstone interval. The development of interlayers has a positive and beneficial effect on the production of horizontal wells, effectively blocking the rise of edge and bottom water.

Table 1 provides few properties for rock and fluids from the reservoir. The original formation pressure in the middle of the L reservoir is 24.56MPa, the saturation pressure is 1.25MPa, the difference between formation pressure and saturation pressure is 23.304MPa, and the pressure coefficient is 1.02, which belongs to a normal pressure system. The formation pressure increases with depth, with a pressure gradient of 0.96 MPa/100m. The temperature of the oil reservoir is 115.44 °C, with a geothermal gradient of 4.05 °C/100m. The crude oil has good properties, including low specific gravity, low viscosity, low sulphur content, and good mobility. The surface crude oil density is 0.829t/m³; dead crude oil viscosity 5.4mPa. s; Freezing point 34 °C; Wax content 8.1%; Sulphur content 0.14%; Asphalt content 0.39%; Formation crude oil density 0.767t/m³; The viscosity of crude oil is 1.471 mPa.s (cP) at downhole condition. The original dissolved gas oil ratio is 8.91m³/m³. The density of the formation water is 1.025g/cm³ to 1.027 g/cm³, the total salinity is 33400mg/l to 35900mg/l, the pH value is 6.01 to 6.14, and it belongs to the CaCl₂ water type.

Table 1—reservoir parameters and fluid properties

Porosity, %	Permeability, mD	reservoir pressure, MPa	reservoir temperature, °C	saturation pressure, MPa	RS, m ³ /m ³	oil density, g/cm ³	Live oil viscosity, mPa.s
20.6	12500	24.2	115	1.25	8.9	0.767	1.5

Advanced Well Completions:

Various challenges, such as early water and gas breakthrough, reduced oil recovery etc. are the result of variation in the reservoir properties, including the fluid properties, petrophysical properties, layer pressure, and fluid contacts, in zones intersected by the horizontal well. These challenges can be mitigated by deploying advanced well completions that can manage the reservoir fluid influx toward the wellbore and

therefore optimise the performance of wells [1-8]. In addition, several studies proved the application of advanced well completions as a type of insurance policy against geological and dynamic reservoir uncertainties [5] that reduces the risk and variation in the expected oil production profiles.

The main components of advanced well completions are down-hole flow controls devices such as Inflow Control Device (ICD), Autonomous Inflow Control Device (AICD) and Inflow Control Valve (ICV) and Annular Flow Isolation (AFI) e.g. swellable packer. ICDs with fixed, simple and cost-effective configurations are passive devices that influence the well's inflow. AICD, as a new generation of the ICD, was introduced to function as a standard ICD prior to the breakthrough (proactive solution) and restrict the production of unwanted effluents with lower viscosity after breakthrough such as gas in light oil and both gas and water in heavy oil production (reactive solution).

There are different technologies that have been commercialized as AICD devices. The most widely deployed AICD is the Rate Controlled Production (RCP) device. Figure 1 illustrates the principal components of the RCP AICD.

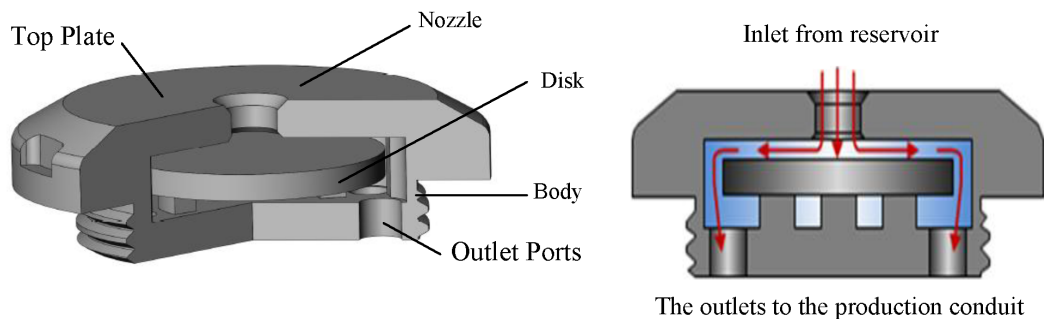


Figure 1—(a) RCP type AICD construction and (b) flow paths.

The RCP AICD is typically incorporated as part of a screen joint as shown in Figure 2. The produced fluids enter the completion through the screen and flow in the annular space between the screen and the unperforated base pipe into the AICD housing where the device is mounted. Fluids then flow through the AICD into the interior of the production conduit where they combine with the flow from other zones.

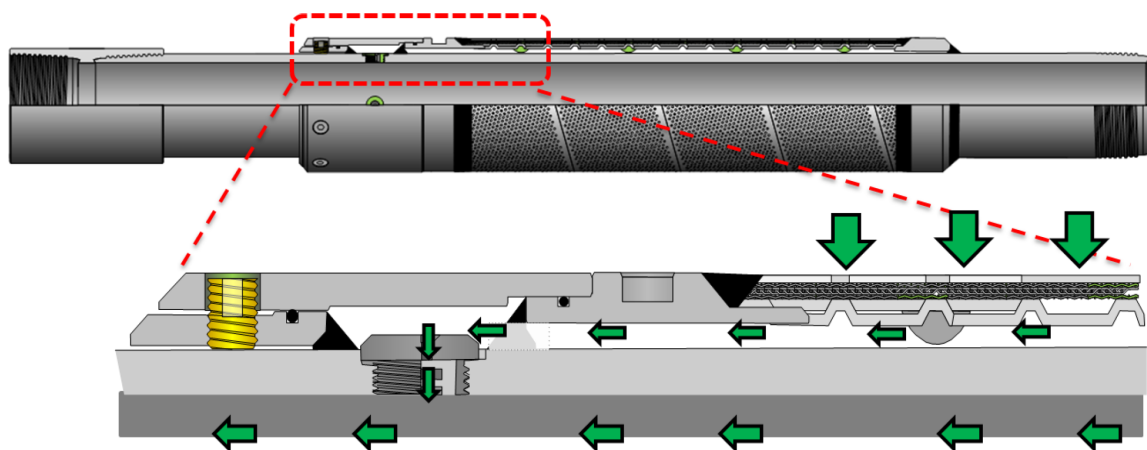


Figure 2—AICD unit mounted into sand screen joints illustrating production flow path.

The AICD generates a variable pressure drop, depending on the rate and fluid properties, based on the size of the inlet nozzle and on the gap created between a levitating disk and the top plate of the housing in which it is contained. Flow enters the device through the nozzle in the top plate of the body, impacts the

disk and spreads radially through the gap between the disk and the top plate, then turns around the disk and is discharged through several outlet ports in the body (Figure 1 b).

Prior to Drilling Study- Well Completion Selection

The basecase completion for the wells were ICD completion. The production data including liquid rate and water (as per Table 2) were used from analogue wells for this study and then the AICD/ICD completion performance were compared under different water cuts from 30 to 90% plus sensitivity at different fluid rates of 4000, 4500, 6000 and 7100 (bbl/d).

Table 2—initial comparison of completion options.

	ICD	AICD	ICD	AICD	ICD/AICD	ICD/AICD	ICD/AICD
Liquid Rate	Water cut	Water cut	Oil Rate	Oil Rate	Water cut difference	Oil rate increment	Annual oil increment
[Sm ³ /day]	%	%	[Sm ³ /day]	[Sm ³ /day]	%	[Sm ³ /day]	[Sm ³]
667	31	30	462	466	0.6	4	1532
672	36	35	428	436	1.2	8	2884
648	42	40	376	388	1.8	11	4145
653	48	45	340	357	2.6	17	6170
636	53	50	301	320	3.1	20	7254
629	65	60	218	249	5.0	31	11419
629	74	70	165	187	3.5	22	8031
722	84	80	116	142	3.5	25	9302
950	81	75	184	236	5.5	52	18942
1130	89	85	125	169	3.9	44	15980

Table 2 shows the difference between ICD/AICD added value is relatively small at lower water cuts while AICD performs significantly better than ICD under conditions of medium to high water cut and higher liquid rate, with reduction up to 5% in water cut.

As the well is expected to be under later conditions i.e. medium to high water cuts for majority of its lifetime, the water control effect of AICD completion technology will be significantly better than that of ICD technology and it will play an irreplaceable role on the well performance. In addition, due to the significant uncertainty in geological structure and the low height of the oil column, AICD was decided to be installed in the well for water control completion to prevent rapid coning of both edge and bottom water.

AICD Completion Design and Performance

As mentioned earlier, well LH1, which was currently in a temporary abandoned state, was used to sidetrack to drill a horizontal well to improve the well network in the northwest area of the oilfield, increase the degree of reserve utilization and delay production decline. In order to reduce reservoir risks and facilitate horizontal trajectory control, the planned horizontal section was about 500 meters long, as far away as possible from the oil-water contact. The planned well has 5135.0 m MD, 2532.3m TVD and -2475.8 m TVDSS. However, as shown in Figure 3, the actual completed drilling depth reached 5299 m MD, 2528.3m TVD and -2471.8 m TVDSS with the length of horizontal section of 918.0 m and the net sand length of 888.0 m. The data from the well confirms that the reservoir section is mainly composed of fine sandstone, with stable lithology, relatively good physical properties and minor limestone content, dark yellow under fluorescent direct illumination, milky white under drip illumination, with a maximum TG value of 1.45%,

all components present. The resistivity value ranges from 11.294 to 40.051 ohm. m, which can be interpreted as an oil layer, totalling 888.0m.

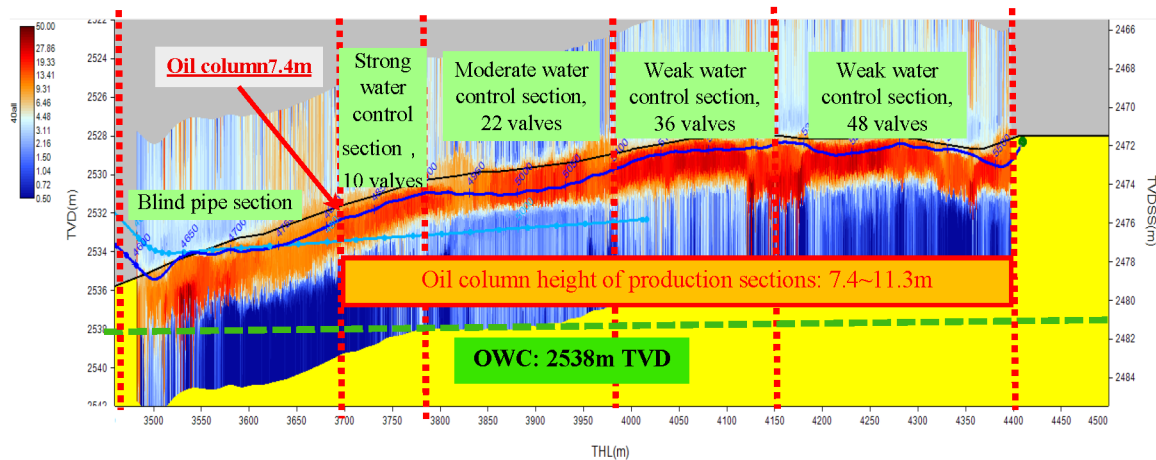


Figure 3—LH1 well geo-steering and AICD installation.

Based on the differences in physical properties of well logging interpretation results and the height difference between the actual drilling trajectory and the oil-water contact, a strategy of strong water control in the low end well section and weak water control in the high end well section is used for completion design. The final specific plan for AICD water control is shown in the figure above:

1. AICD water control uses a total of 116 devices to meet maximum expected liquid rate of 2385m³/d (15 kstbpd) from the well.
2. Four swellable packers divide the entire horizontal well section into 5 sections. It is necessary to segmentize the well into separate sections along highly heterogeneous formations such as formation L, to minimise interaction between zones with varying physical properties. This includes optimizing the length of each section and the number of packers used as deploying multiple packers enhances overall water control effectiveness significantly. Note that compare to industry standard, relatively fewer packer than optimum number was used in this well, hence there was room for further improvement.
3. Since the first section is located at the deepest point of the horizontal section with proximity to the oil-water contact, it was decided to install blank pipes in this section to prevent being exposed to massive water production prematurely so improve water control and delay water breakthrough time.
4. The second and third sections are in the upward climbing phase of the well trajectory and are relatively low, so it was ultimately decided to install a single valve for a single screen for strong water control in these two sections.
5. The fourth and fifth layers are located at the shallowest section of the anticline structure, farthest away from the oil-water contact and are expected to be exposed to the least risk of water production with optimal crude oil production sections. Dual valves on a screen were used for all screens of both sections to maximise the capacity.

The well has been in production since Jan 2020. Figure 4 shows the well performance over 5 years of production while Figure 5 focused on almost two year (from April 21 till March 2023) when the liquid rate from the well was increased from 8K to over 13kstbd. This increase was delivered by adjusting pump setting and well head choke setting as shown in Figure 6. Figure 6 also shows oil rate increases from ~500 to ~900 stbd while water cut decreases from 95.2 to 93.5% and stabilizes at the level within ~2 yrs. This has been achieved as the result of equipping AICDs to function under higher differential pressure to choke the higher water cut zones more significantly while maximising the drawdown from more oil saturated zones.

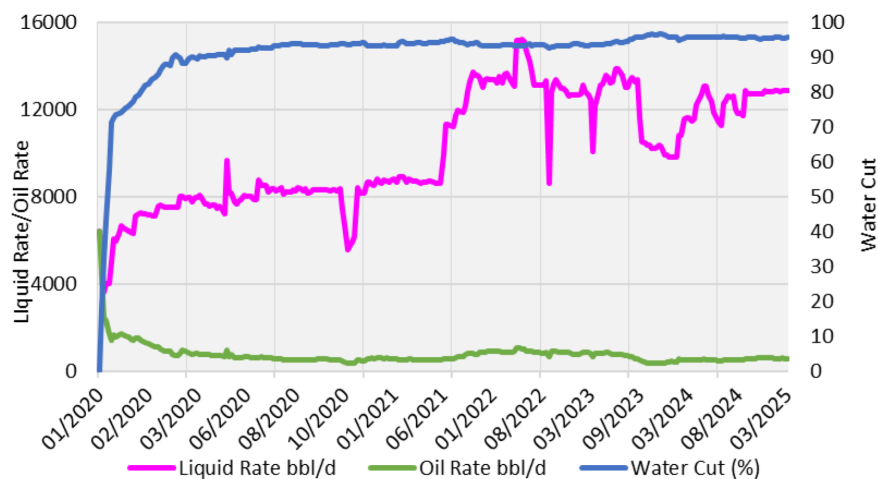


Figure 4—Production data from well from start to April 2025.

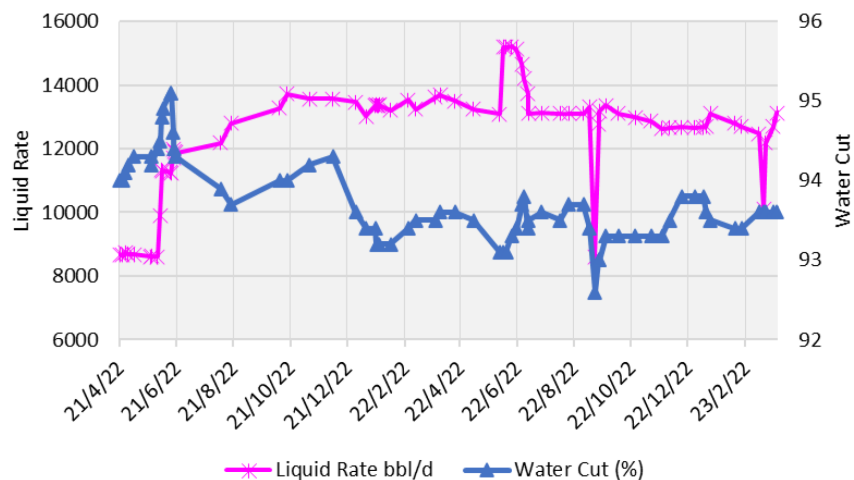


Figure 5—Liquid rate and water cut for period of production, from April 21 till March 2023, when liquid rate was increased.

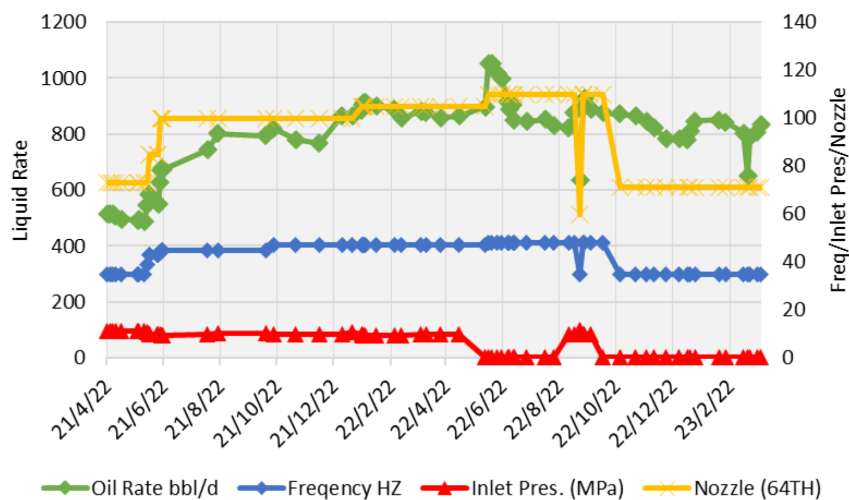


Figure 6—Oil rate, pump and well head settings for production period from April 21 till Mar 2023.

Predicated Performance vs. Reality

ICD has been the basecase completion for the wells in this reservoir prior to this installation. As mentioned, based on real time data from drilling, it was decided that 5 completion zones with 4 packers will be installed in the well.

Figures 7–10 show the actual and predicted oil production rates for the well over the well life. As shown, the well performance with AICDs exceeds expectation from the well with ICDs not only on oil production rate as per Figure 7 but also on the lifetime of well. The well with AICD completion has managed, so far, to deliver 310% increased cumulative oil from the well as per Figure 8 and Figure 9 while extending the well life twice as much than expected by the ICD completion initially. Comparing cumulative oil productions at same production period of 3.5 years, cumulative oil production by AICDs is already 2.2 times more than expected from the ICD well. As shown in Figure 10, the initial produced liquid rate is much higher than the prediction. This resulted in slightly higher initial water cuts than predicted earlier, however, the mid- and long-term water cut has been stabilized at much lower (~93.5% and later 95%) than the common WC of 98 and 99.5% from the analogue wells equipped by ICDs for mid and long terms.

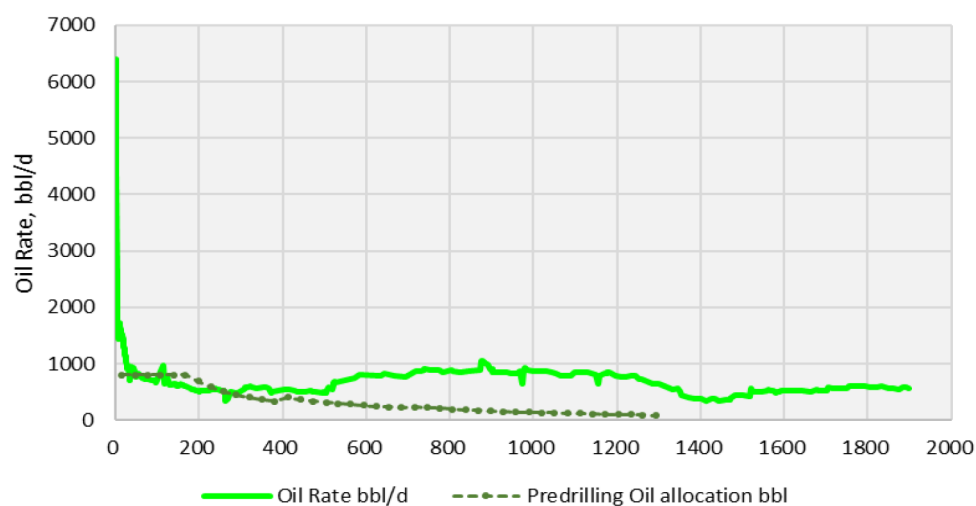


Figure 7—Actual oil rate vs predrilling oil allocation rate.

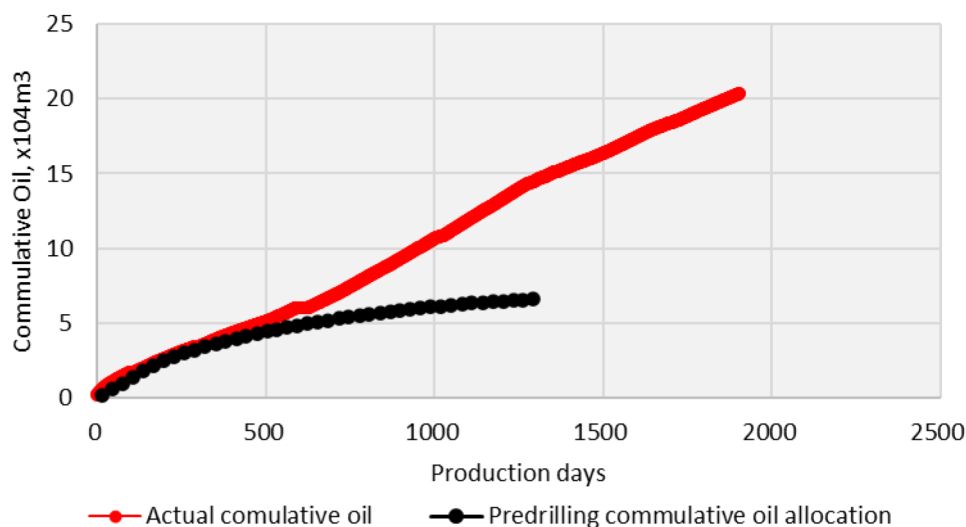


Figure 8—Actual cumulative oil vs predrilling cumulative oil allocation.

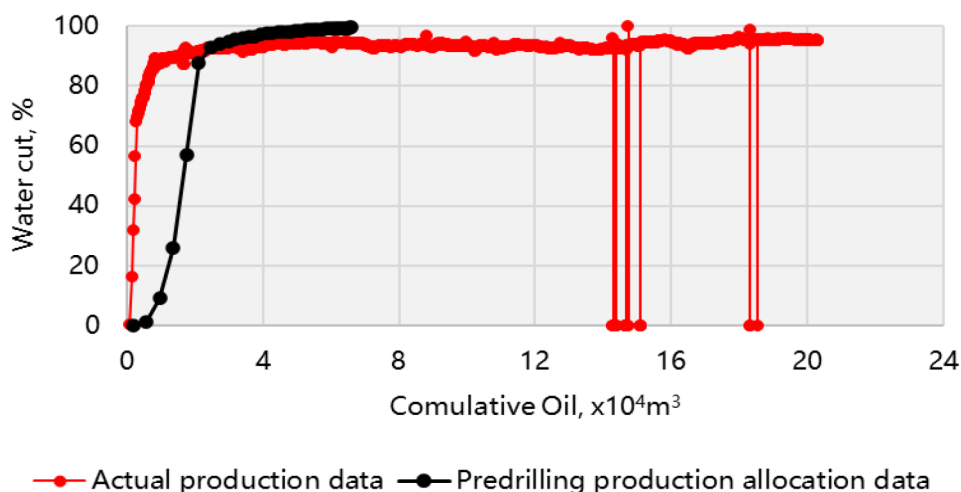


Figure 9—Cumulative actual and predrilling-allocation production versus water cut.

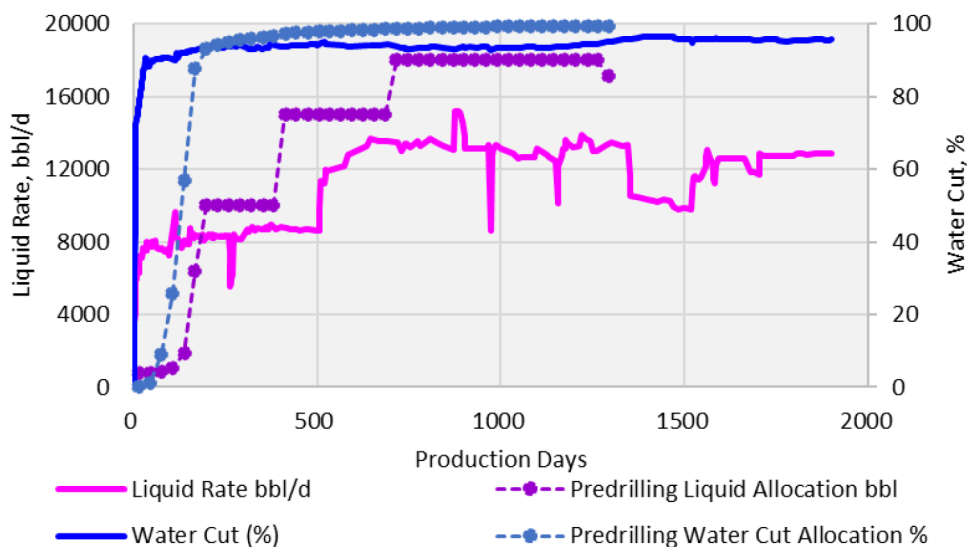


Figure 10—Actual production vs predrilling production allocation.

Summary

This case study demonstrates the effectiveness of AICD technology in controlling water production in edge and bottom water reservoirs with low-viscosity crude oil. The well with AICD completion has managed, so far, to deliver 310% increased cumulative oil from the well while extending the well life twice as much than expected by the ICD completion.

Notably, the crude oil in this well has a low viscosity of 1.47 cP, setting a new world record for the lowest formation crude oil viscosity successfully managed using AICD in oil wells globally to only control water. The results demonstrates that AICD technology is effective not only for water control in mid to high viscosity oils as previously illustrated but also demonstrates excellent water control performance in reservoirs with low-viscosity crude oil.

The number of deployed devices was selected to allow max. production of 15 kstbpd. However, at early stages, the well was on much lower production rate due to the surface limitations. However, by increasing the liquid rate later on March 2021, thereby raising the pressure differential across the AICD valves, it becomes possible to restrict flow from zones with high water cut. Simultaneously, this allows greater production from zones with higher oil saturation resulting in lower water cut.

This strategy ultimately helps to reduce the overall water cut, enhance oil production and improve the field's recovery efficiency. AICD mitigates water rate production and reducing operating costs for lifting, processing and disposal of water plus greenhouse gas emission and energy consumption associated with the operations as being evaluated for similar objectives as presented on [6&8].

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